

Stage-1 Calibration Captures

Collecting sphere and board data to correct the depth sensor's range and normals



The session produces data plus an exact table of poses



186

robot poses, 5 frames each



~2 h

robot time with fixture changes and checks



0.1 mm

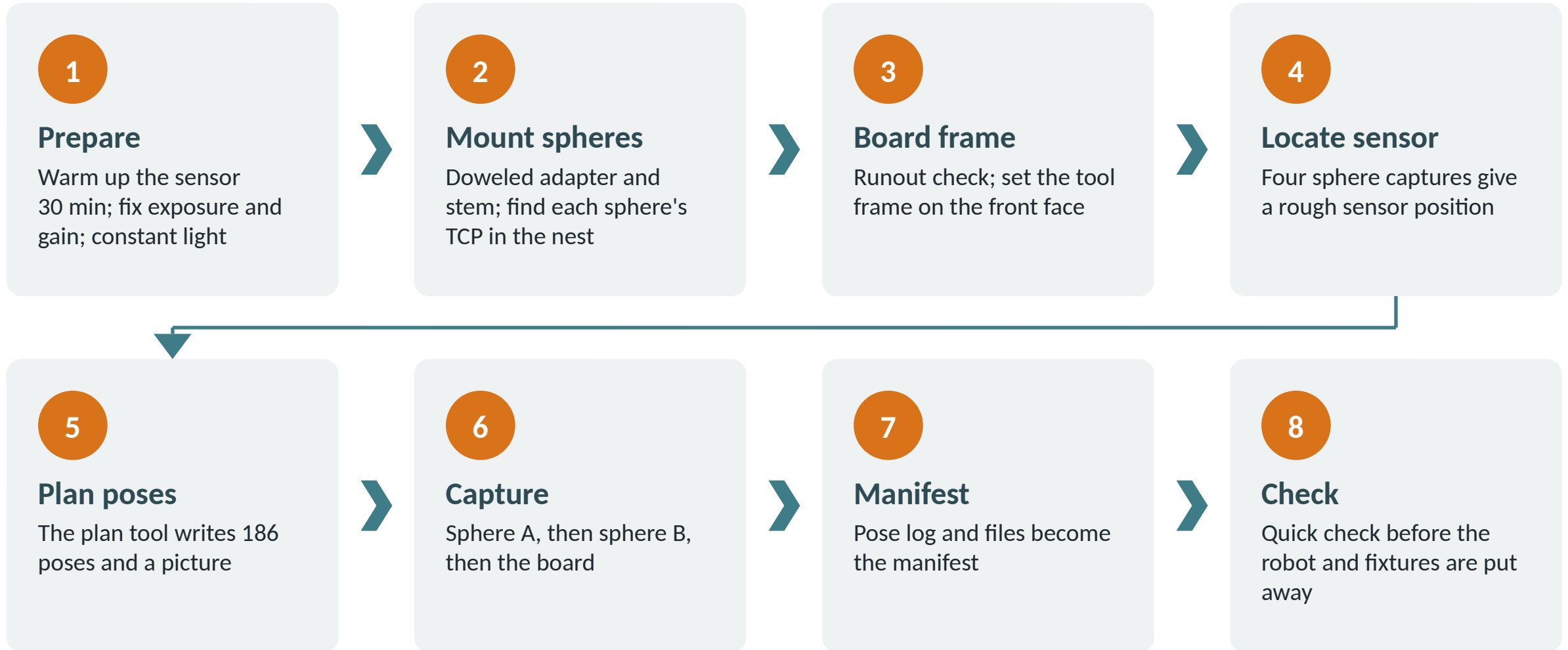
accuracy target at 500 mm, scaling with distance



If the pose table is wrong, the calibration is wrong.

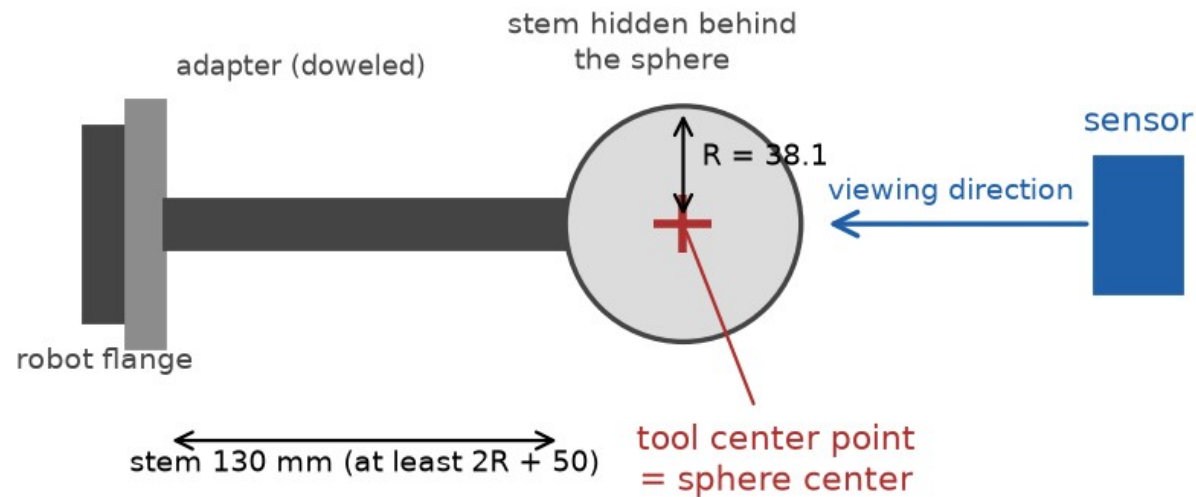
Every capture file must be matched to the target that was in view and to exactly where the robot put it. Most of the procedure is about getting that table right.

The session at a glance



Two spheres and a board carry the known geometry

(a) sphere on its stem, side view



Sphere on its stem, side view: the stem points away from the sensor so that the sphere hides it.



Sphere A

76.2 mm diameter; used from 300 to 550 mm



Sphere B

152.4 mm diameter; used from 500 to 1,100 mm



Board

200 x 150 mm, at least 6 mm thick, flat to 0.05 mm after painting; tilted 0, 20 and 40 degrees

New equipment costs about \$1,900 to \$5,400

Item	Estimated cost (USD)
Sphere A, bearing ball, bonded stem, blasted, measured	\$350 to \$850
Sphere B, turned aluminum, anodized, measured	\$600 to \$1,800
Stems, two	\$150 to \$400
Adapter plates, two	\$250 to \$600
Board, glass or ground aluminum, painted	\$50 to \$400
Board adapter	\$250 to \$600
Three-ball nest	\$150 to \$450
Dial indicator with magnetic base	\$100 to \$300

\$1,900 to \$5,400

bearing-ball sphere A, aluminum sphere B

\$3,550 to \$10,250

both spheres made to order

Estimates of October 2026, plus or minus 50 percent; sphere B is the least certain. The sensor, robot and capture computer are in hand.

Spheres must be matte and measured, not certified



Finish

- Every bearing ball arrives mirror-bright: unusable
- Bead-blast with fine glass bead: removes micrometers
- Matte paint is second choice: 0.02 to 0.05 mm per coat
- "Chrome steel" is the 52100 alloy, not plating



Diameter

- Known to 0.025 mm, measured after the finish
- Micrometer at several orientations, or a CMM
- Aluminum grows 0.0035 mm per degree C: measure at session temperature
- The measured value goes in the pose log

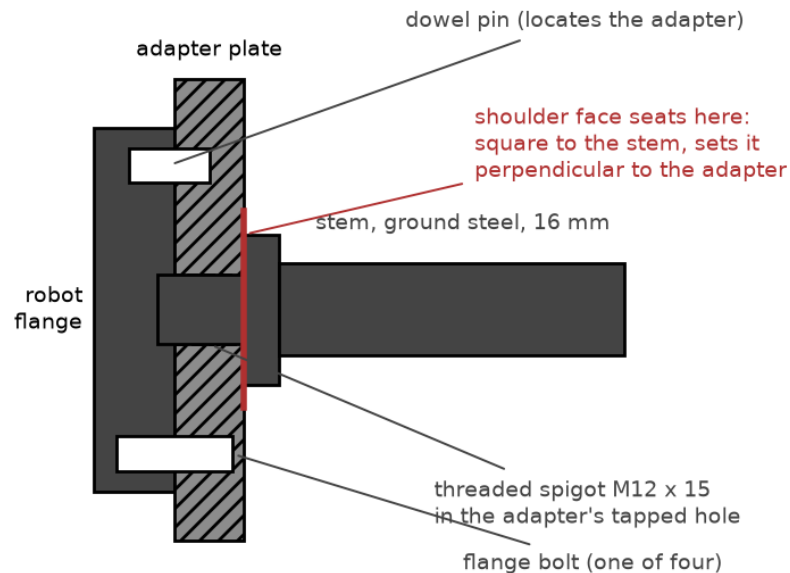


Mounting

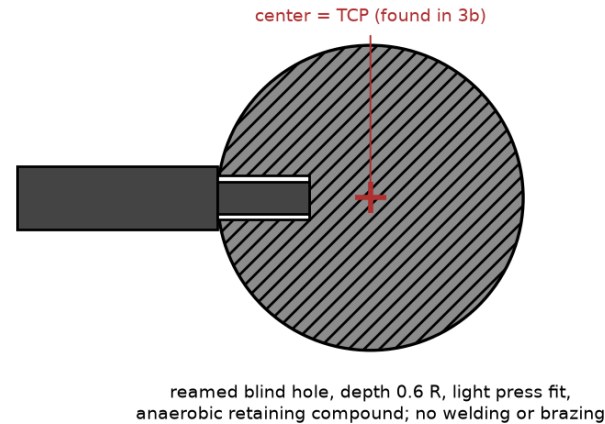
- Hardened balls cannot be tapped
- Steel ball: drilled or EDM blind hole, bonded stem
- Aluminum sphere B: thread cut when it is turned
- No pre-tapped precision sphere exists above 1.5 inch

A stiff, repeatable chain from flange to sphere

(a) flange, adapter and stem shoulder, in section



(c) steel sphere: bonded blind hole



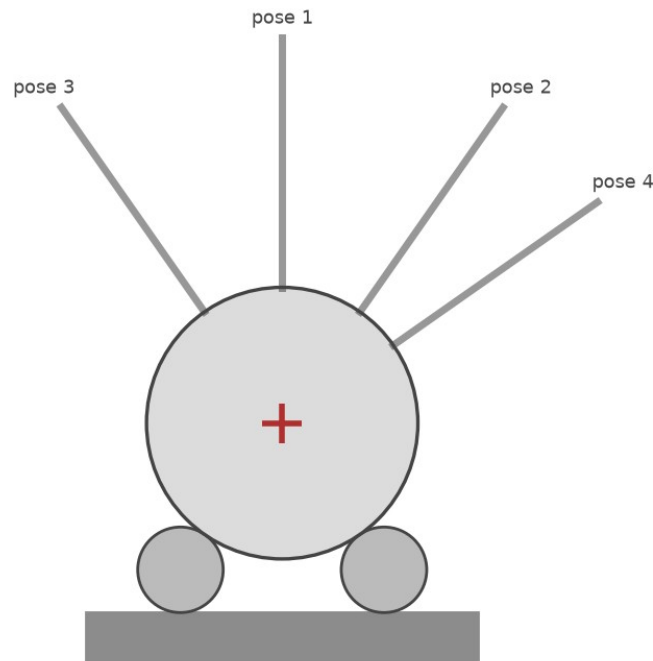
Left: adapter on the flange's dowel, stem shoulder seated on the adapter face. Right: steel sphere with a bonded blind hole.

- Adapter located by the flange dowel, so a remount lands in the same place
- Ground stem, 16 mm for A and 30 mm for B, keeps sag to 0.02 to 0.05 mm
- Stem length at least $2R + 50$ mm, so the sphere hides the stem
- Witness mark across stem and adapter shows a loosened joint
- After any remount, re-check in the nest: readings within 0.1 mm

The three-ball nest finds each sphere's center

(b) finding the tool center point

the center stays put while the wrist turns:
the robot's TCP routine solves for that point



three-ball nest (two balls visible), bolted down

1

Enter a rough TCP from caliper measurements

2

Run the robot's multi-orientation TCP routine with at least six seatings: stem up, tilted 40 degrees four ways, one twisted

3

Accept only a routine error of 0.1 mm or less

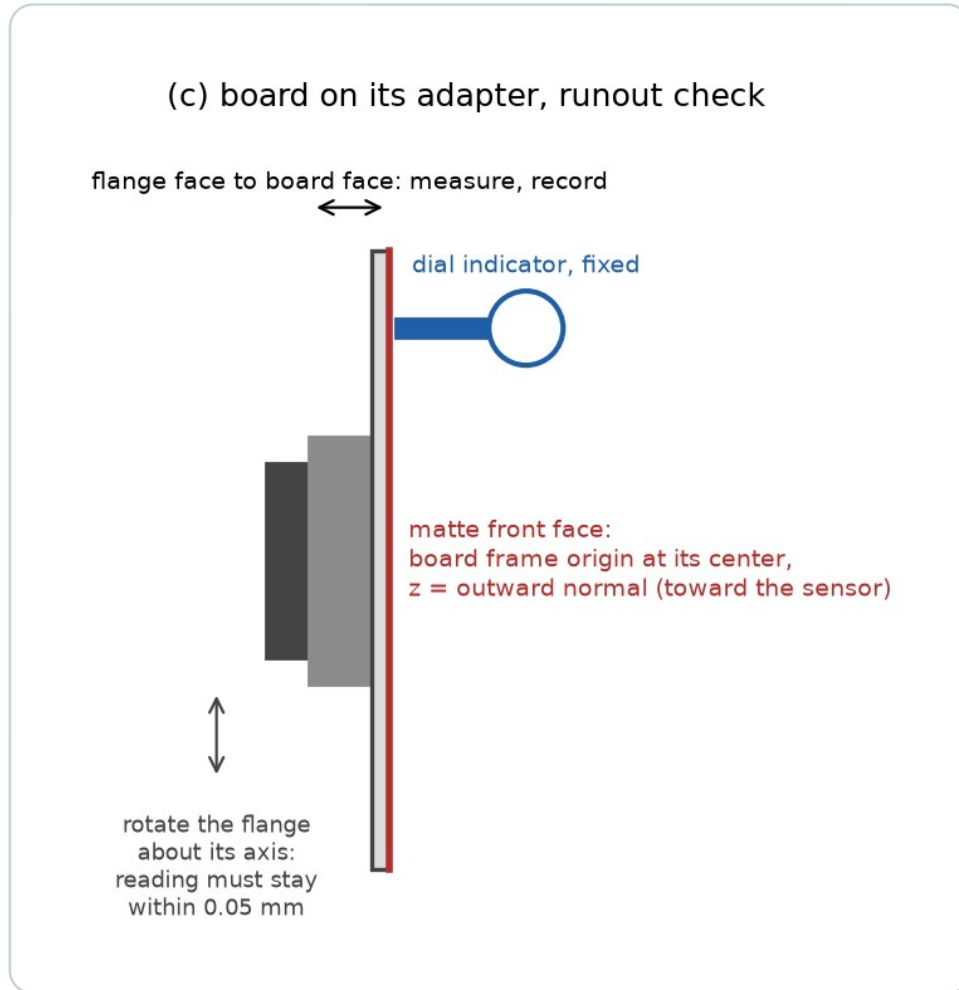
4

Re-seat at a very different wrist angle: positions agree within 0.1 mm

5

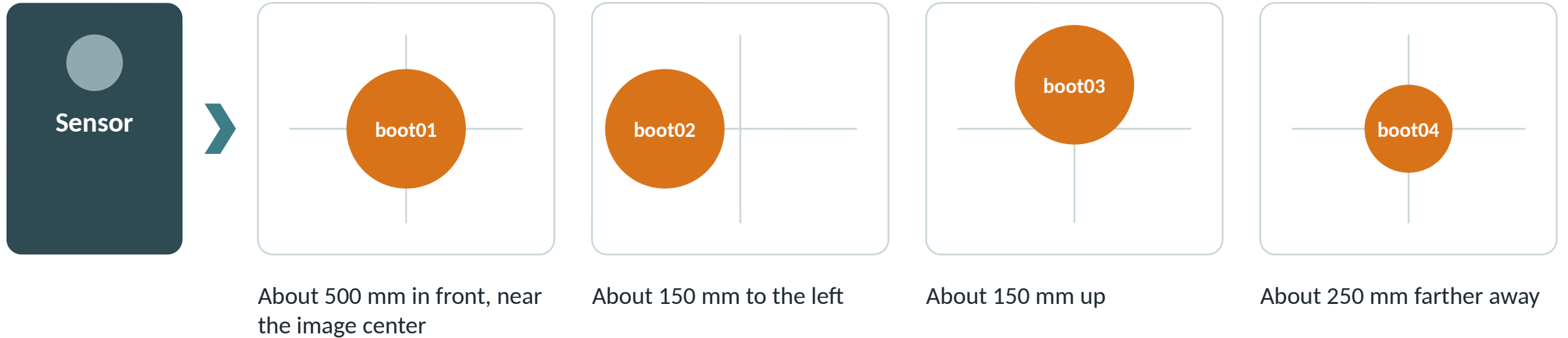
Save TCP_SPHERE_A_76mm and TCP_SPHERE_B_152mm

The board's tool frame sits on its front face



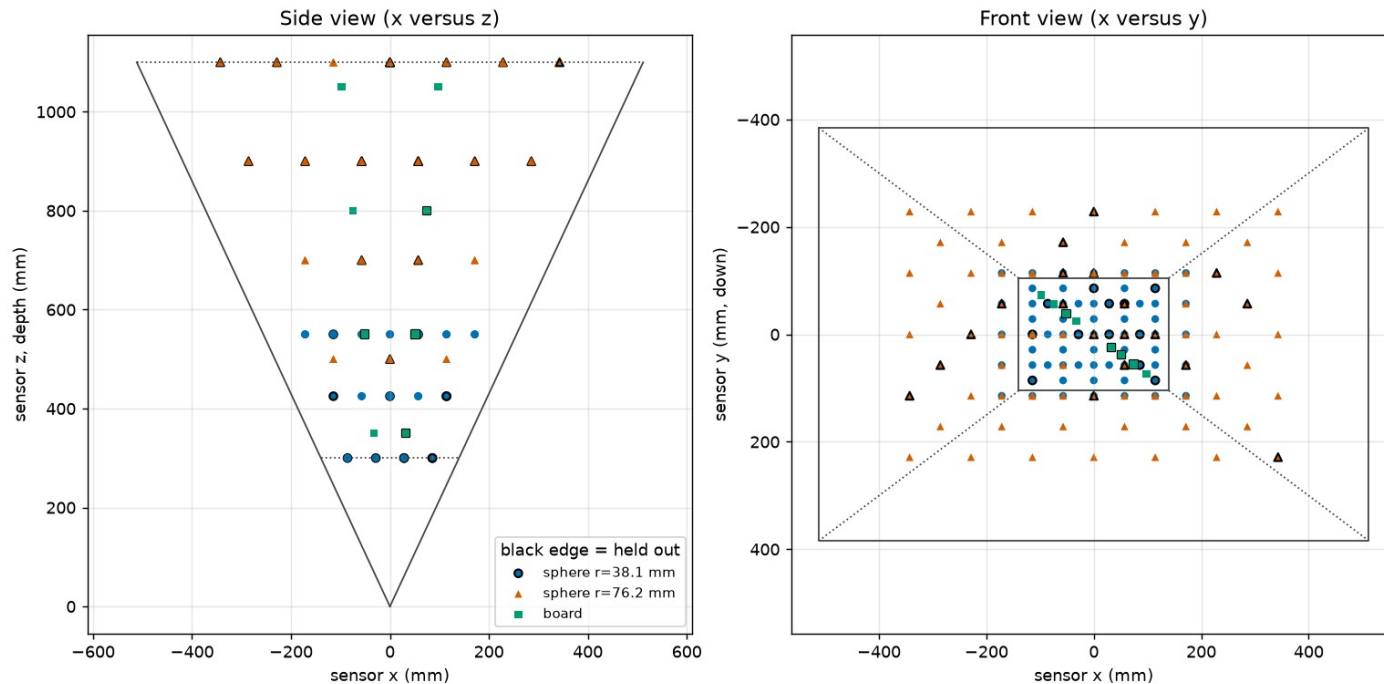
- ✓ Runout: rotate joint 6 through 360 degrees; the dial indicator stays within 0.05 mm
- ✓ Measure flange-to-face distance D at four places; they agree within 0.05 mm
- ✓ Tool frame: origin at the face center, z out of the face, x along the long edge
- ✓ A few degrees of error in x is harmless; an error in z is not
- ✓ Save as TOOL_BOARD; record D, the runout and the board size

Four captures locate the sensor well enough to plan



- Record the robot's reported TCP position for each
- The check tool fits the sphere centers; the plan tool fits the sensor position
- Residuals of a few millimeters are fine; tens of millimeters mean a copying error or a wrong TCP

The plan spreads 186 poses through the volume



About 20 percent of poses are held out to test the fit; they are captured like all the others.

90

sphere A poses at 300, 425 and 550 mm

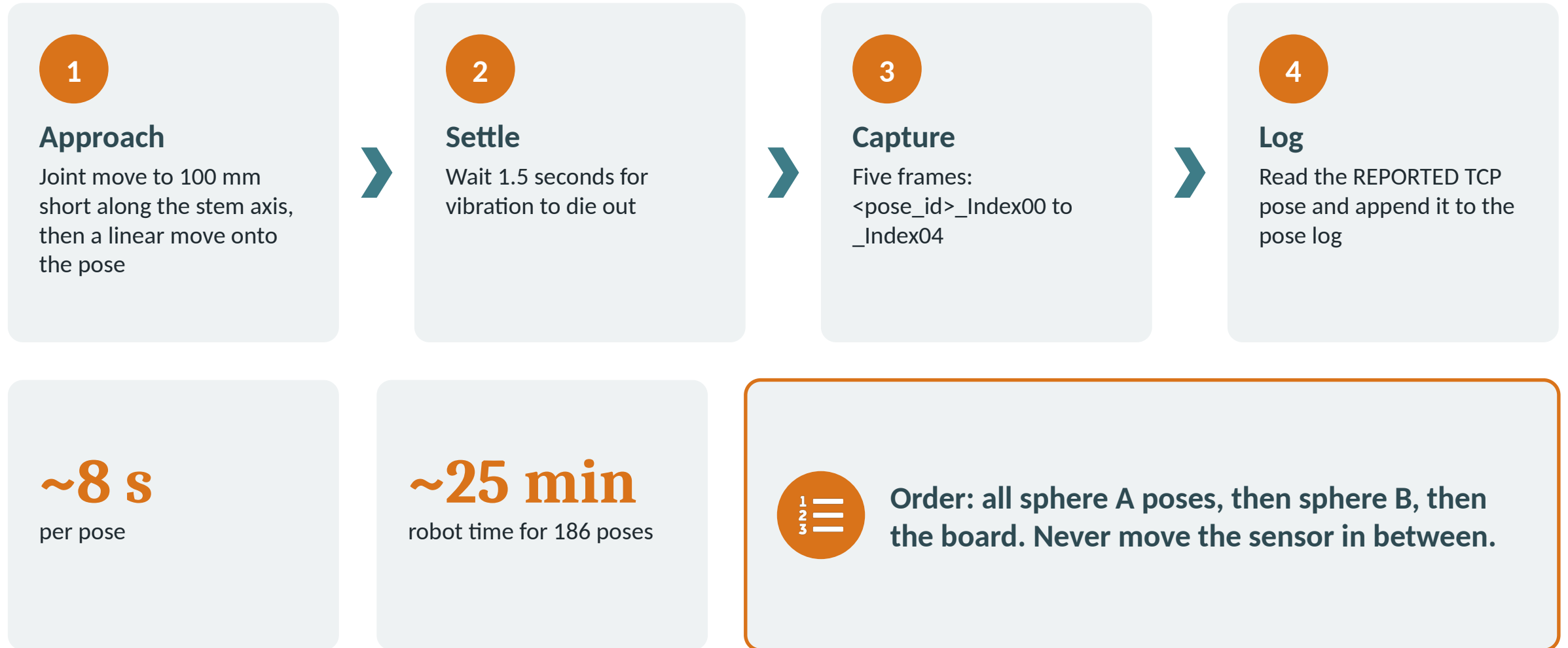
57

sphere B poses from 500 to 1,100 mm

39

board poses at 4 depths, tilted 0, 20 and 40 degrees

Every pose runs the same short robot loop



Poses are recorded in the file header or in a pose log

Option A: in the file header

- The capture software writes robotPose, a 4 x 4 matrix, tool to base
- No separate log needed beyond the target sizes
- Use it if the capture software supports it

Option B: pose log and manifest

- The robot program appends one CSV line per pose
- Orientation in the controller's own form: KUKA A, B, C or FANUC W, P, R, quaternion, rotation vector, matrix
- make_manifest matches files to poses and names every mismatch
- Recommended as a backup even with option A



s038_z0425_017, sphere, 38.10, , , 812.40, -33.21, 455.02, none, , , ,

Check the captures before the robot is put away



Low valid fraction

below 50 percent of frames

Target not read: check exposure, or the target was outside the field



Touches the image border

valid pixels at the edge

Target partly out of view: the pose is unusable; re-plan it slightly inward



Sphere center off

more than 2 mm

One pose: position copied wrongly. One sphere: wrong TCP. Growing across the volume: base frame or robot accuracy



Board normal off

more than 2 degrees

Board tool frame orientation wrong, or the rotation type in the log is misnamed

Re-capture a flagged pose under a new id, for example s038_z0425_017r; never delete lines from the log.

Without a ball bar, two shape checks stand in



Sphere shape

- Each held-out sphere is fitted with its radius free
- After correction the fitted radius should match the measured radius
- Judge each error against its standard error, which is large for a small or distant sphere



Board flatness

- Each held-out board is fitted with its own plane
- After correction the spread should fall toward the plate's flatness
- Sees only errors that bend the board; mainly catches the fixed pattern



Neither check tests the scale of the volume or a smooth robot error; those rest on the robot's 0.1 mm accuracy.

Seven things that spoil a session



Moving or bumping the sensor: everything after it is a new session



Fingerprints or gloss: wipe the targets with isopropyl alcohol



Changing exposure, gain or the base frame mid-session



Letting the stem point toward the sensor: the sphere must hide it



Logging the commanded pose instead of the reported pose



Mixing up the two spheres' radii in the pose log



A loose adapter or a bent stem: roll the stem on a flat surface first

What to hand over at the end of the session

- ✓ captures/ with every .mc file, named <pose_id>_Index<nn>.mc
- ✓ pose_log.csv, or confirmation that the headers carry robotPose
- ✓ captures/manifest.csv, built without errors
- ✓ captures/check.json with no flags, or a note on each
- ✓ plan/ files as used, boot.json and the bootstrap captures
- ✓ session_notes.txt: sensor serial, warm-up, exposure and gain, base frame, TCPs, board D and runout, sphere diameters, temperature
- ✓ Photos of the sensor mount, each fixture on the flange, and the nest



Software

- plan_poses, make_manifest and check_captures in sphcal/cli
- Install and self-test: procedure appendix C
- Every tool prints its options with --help

Open items and next steps

Open

- ? Sphere procurement awaits a canvass of the market
- ? Candidate provisional far-range sphere B: Rothbucher RSLB10M, 145 mm, sphericity 0.3 mm

Decided

- ✓ D-13: sphere diameters known to 0.025 mm after finishing
- ✓ D-14: no ball bar; shape checks stand in

Next steps

- 1 Canvass the sphere market and order
- 2 Machine the adapters, stems, board and nest
- 3 Dry run: TCPs, board frame, bootstrap
- 4 Stage-1 session, about two hours
- 5 Fit, read the held-out results, decide on stage 2